BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . WEEK 4 . MODULE 6 OF 16

Module 6: Muscular System

Topics: Skeletal Muscle Microanatomy, Sliding Filament and the Cross-Bridge Cycle, Motor Units and Muscle Mechanics

TOPIC 1 OF 3 . WEEK 4 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Skeletal Muscle Microanatomy

From whole muscle down to the sarcomere: the structural ladder of contractile tissue.

The Science

The structural hierarchy is the spine of the topic: muscle, fascicle, fiber (cell), myofibril, sarcomere. Each level is wrapped in connective tissue (epimysium, perimysium, endomysium) that converges into the tendon. Build this scaffold first; everything else hangs on it.

The sarcomere is the contractile unit. Z-line to Z-line. Thick filaments (myosin) in the center (A band). Thin filaments (actin, troponin, tropomyosin) extending from the Z-line (I band). The overlap region is where cross-bridges can form. During contraction the I band shrinks and the H zone (myosin only) narrows, but the A band (thick filament length) does not change. This is a common test point and a useful conceptual anchor.

Excitation machinery: sarcolemma (the muscle fiber's plasma membrane), T-tubules (deep invaginations that carry action potentials into the interior), sarcoplasmic reticulum (SR, the calcium store wrapping each myofibril). T-tubules and SR meet at triads, where electrical excitation triggers calcium release. This coupling is essential; disrupt the triads and contraction fails even though action potentials still propagate.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the structural hierarchy as nested levels, then a sarcomere with bands labeled. Pre-commitment makes the next topic (sliding filament) much easier.

Common misconceptions to address

- Students think the A band changes length during contraction. It does not; the thick filament is rigid.
- T-tubules and SR are confused. T-tubules carry action potentials inward; SR stores and releases calcium. Both meet at the triad.
- Sarcomere is sometimes called sarcomeres (plural). Get the singular cue: it is one unit, repeated.

Order of operations (what to teach first)

Hierarchy first, then sarcomere structure, then excitation machinery. Save the cross-bridge cycle for the next topic; it is its own beat.

Self-test prompts

- Without notes, list the structural levels from whole muscle down to sarcomere.
- Sketch a sarcomere and label A band, I band, H zone, Z-line.
- What is the function of a triad?

Why This Matters to Your Patients

Muscular dystrophies are diseases of structural proteins (dystrophin in Duchenne). The sarcolemma cannot withstand the mechanical stress of contraction, fibers break down, and progressive weakness results. Malignant hyperthermia (next topic) involves a defective SR calcium channel. Statins occasionally cause myopathy: muscle pain and weakness from sarcomere-level disturbance. Understanding skeletal muscle microanatomy is understanding why a patient with myopathy is weak and what tests confirm it (CK elevation, biopsy).

TOPIC 2 OF 3 . WEEK 4 . 10 CARDS (5 DOK 1 / 3 DOK 2 / 2 DOK 3)

Sliding Filament and the Cross-Bridge Cycle

How calcium, ATP, actin, and myosin convert chemical energy into mechanical force.

The Science

This is one of the most beautiful pieces of physiology and one of the highest-yield topics in A&P. Spend the time.

Step 0 (the trigger): action potential at the neuromuscular junction releases ACh, opens nicotinic receptors, depolarizes the sarcolemma. The depolarization sweeps down T-tubules and triggers calcium release from the SR. Calcium is the signal.

The four-step cross-bridge cycle: (1) Cocking, myosin head hydrolyzes ATP to ADP+Pi and reorients into the high-energy position. (2) Binding, calcium binds troponin, tropomyosin shifts off the actin sites, myosin binds actin. (3) Power stroke, myosin pivots, pulling actin toward the M-line, releasing ADP and Pi. (4) Detachment, a new ATP binds myosin, which releases actin; cycle repeats. The thin filaments slide along the thick filaments; the sarcomere shortens; the muscle contracts. Filament lengths do not change; their overlap does.

Relaxation: when neural input stops, ACh is broken down by acetylcholinesterase, the sarcolemma repolarizes, the SR calcium ATPase pumps calcium back into the SR, troponin returns to its resting position, tropomyosin re-covers the actin sites. No calcium, no cross-bridges, no contraction. Both excitation and relaxation require ATP. This is why rigor mortis happens: after death, no ATP means myosin cannot detach from actin.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a sarcomere at rest and predict what changes during contraction. Then they have to draw the four-step cycle in order before watching.

Common misconceptions to address

- Students think calcium directly attaches myosin to actin. Calcium binds troponin; troponin moves tropomyosin; tropomyosin exposes the actin site.
- ATP is thought to be needed only for the power stroke. It is needed for cocking AND detachment, but the power stroke itself is the release of stored energy.
- Rigor mortis is sometimes thought to be from low calcium. It is from no ATP, so no detachment, so locked cross-bridges.

Order of operations (what to teach first)

Trigger first (ACh to calcium), then the four steps drilled IN ORDER with hand gestures, then relaxation, then rigor as a check on understanding.

Self-test prompts

- Without notes, list the four steps of the cross-bridge cycle in order.
- Where in the cycle is ATP hydrolyzed? Where is fresh ATP needed?
- Explain rigor mortis in terms of cross-bridge biology.

Why This Matters to Your Patients

Curare and similar drugs block nicotinic ACh receptors at the neuromuscular junction, preventing depolarization, causing flaccid paralysis. Used clinically for surgical muscle relaxation. Organophosphate poisoning inhibits acetylcholinesterase, so ACh lingers, producing fasciculations and eventually depolarization block and respiratory paralysis (the farm worker's emergency, the sarin gas attack). Malignant hyperthermia is a genetic defect in the SR calcium release channel: certain anesthetics trigger uncontrolled calcium release, sustained contraction, hyperthermia from continuous cross-bridge cycling, ATP depletion, and rhabdomyolysis. Treatment is dantrolene (blocks SR calcium release). Every one of these stories is one node of the cross-bridge cycle being attacked.

TOPIC 3 OF 3 . WEEK 4 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Motor Units and Muscle Mechanics

Motor unit organization, recruitment, summation, and fatigue.

The Science

A motor unit is one motor neuron plus all the muscle fibers it innervates. Small motor units (a few fibers) give fine control: extraocular muscles have motor units of 5-10 fibers. Large motor units (many fibers) give power: quadriceps motor units have thousands. The size of the motor unit is set by what the muscle needs to do.

Force grading happens two ways. Recruitment: activate more motor units (the Henneman size principle says small units fire first, large units recruited as force demand rises). Frequency summation: fire the same motor unit faster, so twitches overlap. If firing is fast enough, twitches fuse into a sustained, smooth contraction (tetanus). These two mechanisms are how the body smoothly grades force from picking up a feather to deadlifting a barbell.

Fiber types: Type I (slow oxidative, fatigue-resistant, postural muscles), Type IIa (fast oxidative, intermediate), Type IIx (fast glycolytic, powerful but fatigue quickly). Endurance training shifts the population toward Type I characteristics; power training toward Type IIx. Fatigue has multiple causes: substrate depletion (glycogen, ATP), metabolic byproduct accumulation (H^+, Pi), and neural factors (central fatigue).

Teaching This Topic

Before the video (drawing prompt)

Ask students to predict the motor unit size in an eye muscle versus a thigh muscle. They have to commit before the answer.

Common misconceptions to address

- Tetanus the disease is named after tetanus the contraction, which causes the rigid muscles. Students often miss this connection.
- Students think 'recruitment' is about overall fitness. It is the moment-to-moment neural decision to activate more motor units.
- Fast and slow twitch fibers are sometimes thought to be fixed at birth. They are largely genetic but training shifts the population.

Order of operations (what to teach first)

Motor unit definition, then force grading (recruitment and frequency summation), then fiber types and adaptation.

Self-test prompts

- Why are extraocular muscles innervated by very small motor units?
- What is the difference between recruitment and frequency summation?
- Predict the fiber type adaptations in an endurance athlete versus a sprinter.

Why This Matters to Your Patients

Myasthenia gravis is an autoimmune attack on nicotinic ACh receptors at the NMJ, producing fluctuating weakness that worsens with repeated use. Lambert-Eaton syndrome is an attack on the presynaptic voltage-gated calcium channel; the resulting decrease in ACh release produces weakness that improves with repeated activation (opposite of myasthenia). Critical illness myopathy and disuse atrophy occur in ICU patients within days of immobilization. ALS destroys motor neurons themselves, denervating muscles and producing progressive paralysis. Nurses caring for any patient with weakness need to think about what part of the motor system is failing.